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B. Maximum Cost Permutation

time limit per test: 2 seconds

memory limit per test: 256 megabytes

Recall that a permutation of length n is a sequence of length n such that each integer from 1 to n appears in it exactly once.

Let's define the cost of a permutation as the minimum length of its contiguous subsegment (possibly empty) that needs to be sorted so that the entire permutation becomes sorted in ascending order.

You are given an integer array p consisting of integers from 0 to n , where no positive (strictly greater than zero) integer appears more than once. You should replace zeros with integers so that the array p becomes a permutation.

Your task is to calculate the maximum possible cost of the resulting permutation.

Input

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

The first line of each test case contains a single integer n ($1 \leq n \leq 2 \cdot 10^5$).

The second line contains n integers $p_1, p_2, \dots, p_n$ ($0 \leq p_i \leq n$). No positive integer appears more than once in this sequence.

Additional constraint on the input: the sum of n over all test cases doesn't exceed $2 \cdot 10^5$.

Educational Codeforces Round 182 (Rated for Div. 2)

Contest is running

01:58:49

Contestant



→ Submit?

Language: Python 3.13.2 
Almost always, if you send a solution on PyPy, it works much faster

Choose file: Choose File No file chosen

Submit

Output

For each test case, print a single integer — the maximum possible cost of the resulting permutation.

Example

input	Copy
4 5 1 0 4 0 5 3 0 0 0 4 1 2 3 0 3 0 3 2	
output	Copy
3 3 0 2	

Note

In the first example, you can make a permutation $[1, 3, 4, 2, 5]$ with the cost 3, because you have to sort segment $[2, 4]$.

In the second example, you can make a permutation $[2, 3, 1]$ with the cost 3, because you have to sort segment $[1, 3]$.

In the third example, there is only one possible permutation — $[1, 2, 3, 4]$, with the cost 0, because the permutation is already sorted.

In the fourth example, there is only one possible permutation — $[1, 3, 2]$, with the cost 2, because you have to sort segment $[2, 3]$.

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